

Querying Relational Data in DBMS

What is Querying Relational Data?

- Querying relational data means retrieving data from relational database tables.
- Queries are written using Structured Query Language.
- Queries help users:
 - View data
 - Search records
 - Filter information
 - Analyze data

Definition

Query:

A request to retrieve or manipulate data from a database.

Purpose of Querying Relational Data

1. Retrieve required information
2. Filter records
3. Sort data
4. Combine tables
5. Generate reports

Relational Database Concepts

Relation

- A table in database.

Tuple

- A row in a table.

Attribute

- A column in a table.

Example Student Table

Student_ID Name Course Marks

101	Ravi	CSE	85
102	Anu	ECE	90

Basic SQL Query Structure

```
SELECT column_name  
FROM table_name  
WHERE condition;
```

Important SQL Commands for Querying Data

1. SELECT Command

Definition

- Used to retrieve data from a table.

Example

```
SELECT * FROM Student;
```

Important Points

6. Retrieves records
7. * means all columns
8. Most commonly used SQL command

2. WHERE Clause

Definition

- Used to filter records based on condition.

Example

```
SELECT * FROM Student  
WHERE Marks > 80;
```

Important Points

9. Filters unwanted data
10. Uses conditions

3. ORDER BY Clause

Definition

- Sorts data in ascending or descending order.

Example

```
SELECT * FROM Student  
ORDER BY Marks DESC;
```

Important Points

11. ASC → Ascending order
12. DESC → Descending order

4. DISTINCT Keyword

Definition

- Removes duplicate values.

Example

```
SELECT DISTINCT Course FROM Student;
```

Important Points

13. Displays unique values only

14. Reduces repeated output

5. AND Operator

Definition

- Combines multiple conditions.

Example

```
SELECT * FROM Student  
WHERE Course='CSE' AND Marks>80;
```

6. OR Operator

Definition

- Returns records if any condition is true.

Example

```
SELECT * FROM Student  
WHERE Course='CSE' OR Course='ECE';
```

7. LIKE Operator

Definition

- Searches patterns in text.

Example

```
SELECT * FROM Student  
WHERE Name LIKE 'R%';
```

Important Points

15. % represents multiple characters

16. _ represents single character

8. BETWEEN Operator

Definition

- Selects values within a range.

Example

```
SELECT * FROM Student  
WHERE Marks BETWEEN 70 AND 90;
```

9. IN Operator

Definition

- Checks multiple values.

Example

```
SELECT * FROM Student  
WHERE Course IN ('CSE','ECE');
```

Aggregate Functions in Queries

1. COUNT()

Definition

- Counts number of rows.

Example

```
SELECT COUNT(*) FROM Student;
```

2. SUM()

Definition

- Calculates total.

Example

```
SELECT SUM(Marks) FROM Student;
```

3. AVG()

Definition

- Finds average value.

Example

```
SELECT AVG(Marks) FROM Student;
```

4. MAX()

Definition

- Finds highest value.

Example

```
SELECT MAX(Marks) FROM Student;
```

5. MIN()

Definition

- Finds smallest value.

Example

```
SELECT MIN(Marks) FROM Student;
```

Joining Tables

JOIN Operation

Definition

- Combines rows from multiple tables.

Example Tables

Student Table

Student_ID	Dept_ID
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101	D1
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Department Table

Dept_ID Dept_Name

D1 CSE

Query

```
SELECT Student.Student_ID, Department.Dept_Name
FROM Student
JOIN Department
ON Student.Dept_ID = Department.Dept_ID;
```

Types of Joins

17. INNER JOIN

18. LEFT JOIN

19. RIGHT JOIN

20. FULL JOIN

GROUP BY Clause

Definition

- Groups rows with same values.

Example

```
SELECT Course, COUNT(*)
FROM Student
GROUP BY Course;
```

HAVING Clause

Definition

- Filters grouped data.

Example

```
SELECT Course, AVG(Marks)
FROM Student
GROUP BY Course
HAVING AVG(Marks) > 80;
```

Advantages of Querying Relational Data

21. Easy data retrieval
22. Fast searching
23. Better analysis
24. Supports reporting
25. Handles large data efficiently
26. Improves decision making
27. Reduces manual work
28. Supports filtering and sorting
29. Easy data management
30. Improves productivity

Additional Important Points

31. SQL is used for querying databases.
32. Queries retrieve specific information.
33. SELECT is basic query command.
34. WHERE filters records.
35. ORDER BY sorts data.
36. Aggregate functions perform calculations.
37. JOIN combines multiple tables.
38. GROUP BY creates groups.
39. HAVING filters grouped data.
40. Queries improve database usability.

Real-Time Example

College Database

Query

Find students with marks greater than 80.

```
SELECT Name, Marks  
FROM Student  
WHERE Marks > 80;
```

Short Definitions for Exams

Query

A request to retrieve or manipulate data from a database.

Relational Data

Data stored in tables with rows and columns.

One-Mark Important Points

- 41. SQL is used for querying relational data.
- 42. SELECT retrieves data.
- 43. WHERE filters rows.
- 44. ORDER BY sorts data.
- 45. DISTINCT removes duplicates.
- 46. JOIN combines tables.
- 47. COUNT counts rows.
- 48. AVG calculates average.
- 49. GROUP BY creates groups.
- 50. HAVING filters grouped results.